



第1讲 个性化推荐

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目录

1. 个性化推荐概述
2. 经典推荐方法
3. 深度学习推荐
4. 高级推荐方法



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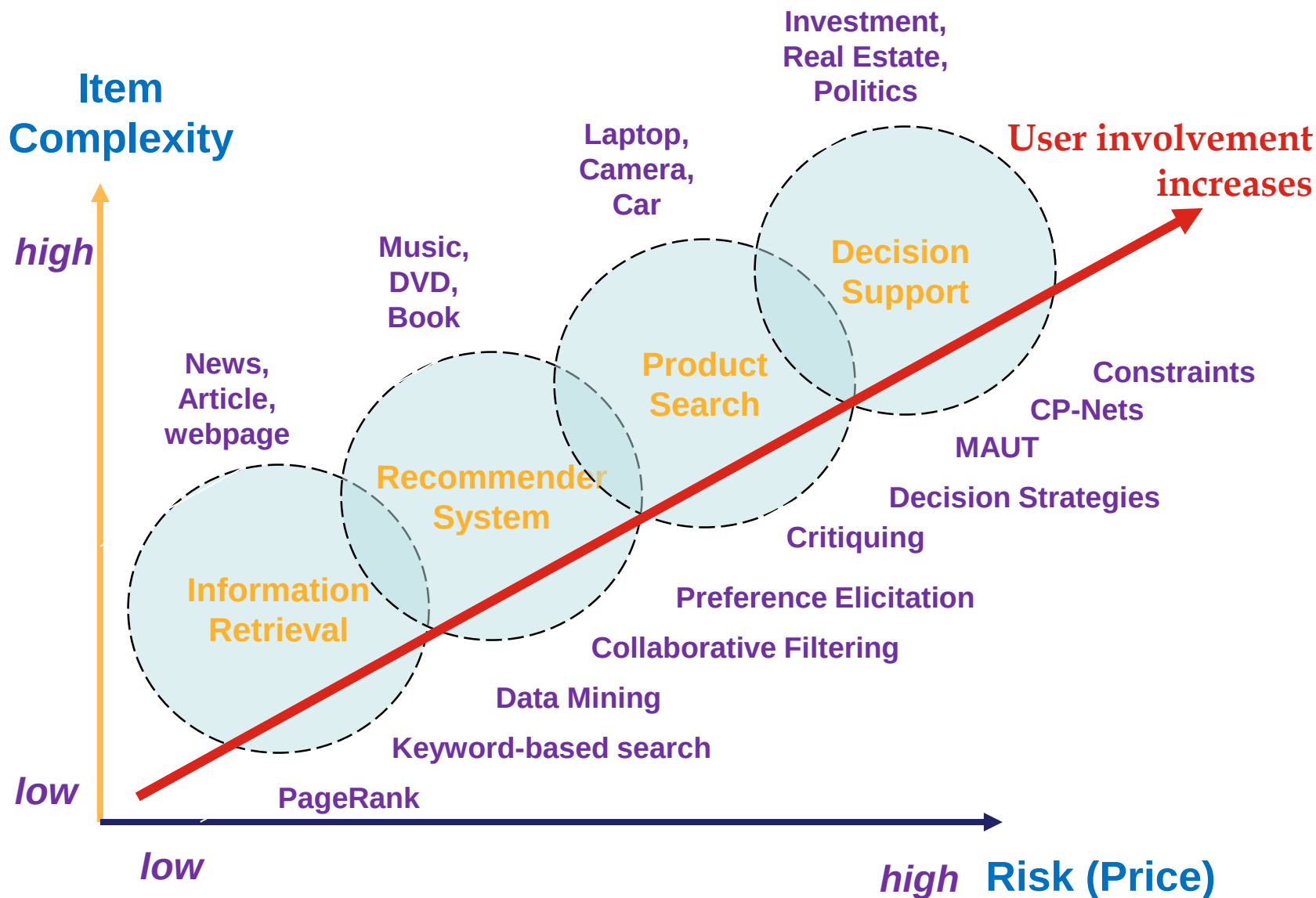


1. 概述

个性化推荐

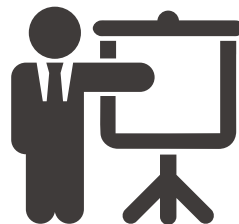


相关方法





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2. 经典推荐方法

个性化推荐的背景



如何实现个性化推荐？

核心技术：个性化推荐方法

基于内容
的推荐

协同过滤
的推荐

基于模型
的推荐

核心思路

你喜欢什么推荐什么

基于内容的推荐

用户
喜欢
什么



这是
什么
商品



用户画像

Item表示

用户（画像）建模

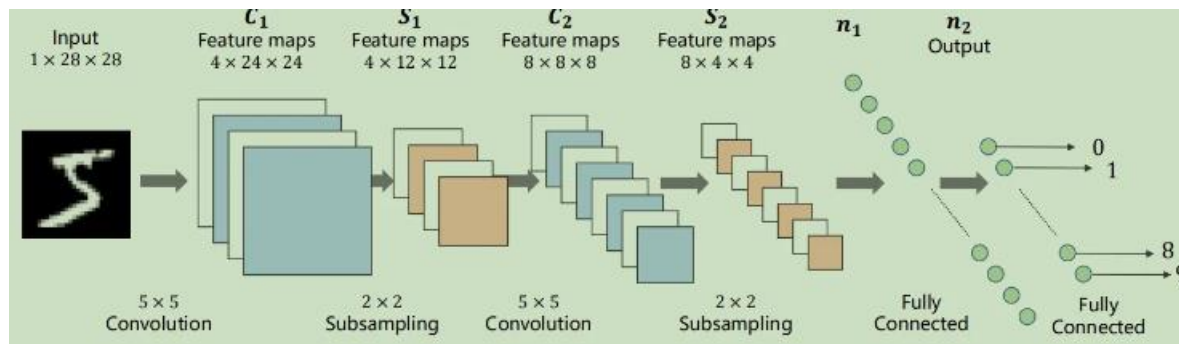
- 个人基本特征属性（**静态**）
- 用户行为大数据（**动态**）
 - 交互大数据
 - 浏览、点击、加入购物车
 - 评论、观看
 - 交易大数据
 - 历史购买
 - 多个平台购买
- 多模态行为数据
 - 表格、文本、视频、音频



数据建模

■ 结构化数据

- 淘宝商品
- 表格型数据



■ 非结构化数据

- 文本数据：新闻、科技论文等
- 网页：HTML、XML
- 多媒体数据：图像、视频、图形、音频

非结构化数据



结构化数据

词袋模型

'it was the worst of times'=[1, 1, 1, 0, 1, 1, 1, 0, 0, 0]

'it was the age of wisdom'=[1, 1, 1, 0, 1, 0, 0, 1, 1, 0]

'it was the age of foolishness'=[1, 1, 1, 0, 1, 0, 0, 1, 0, 1]



TF-IDF模型



计算机科学文档集

“计算机” “科学” “商务”

该文档集中的全部重要词项

- 文档D包含三个词 (a, b, and c):

➤ $\text{freq}(a,D)=3, \text{freq}(b,D)=2, \text{freq}(c,D)=1$

- 假设有10000个文档:

➤ $N_a=50, N_b=1300, N_c=250$

$$d_{t_i} = \left(0.5 + 0.5 \frac{tf_i}{tf_{max}} \right) \left(\log \frac{n}{df_i} \right)$$

- 则对于D文档:

➤ a: $tf = 3/3; idf = \log(10000/50) = 5.3; \quad tf-idf = 5.3$

➤ b: $tf = 2/3; idf = \log(10000/1300) = 2.0; tf-idf = 1.3$

➤ c: $tf = 1/3; idf = \log(10000/250) = 3.7; \quad tf-idf = 1.2$

文本向量空间模型

“乔布斯离我们而去了。”

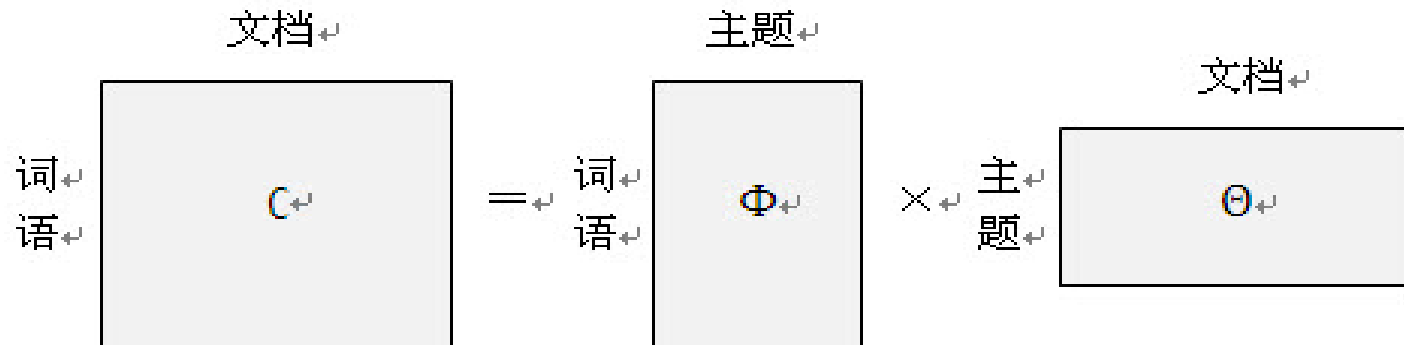
“苹果价格会不会降？”



语义主题挖掘!!!

LDA(Latent Dirichlet Allocation)

$$p(\text{词语} | \text{文档}) = \sum_{\text{主题}} p(\text{词语} | \text{主题}) \times p(\text{主题} | \text{文档})$$

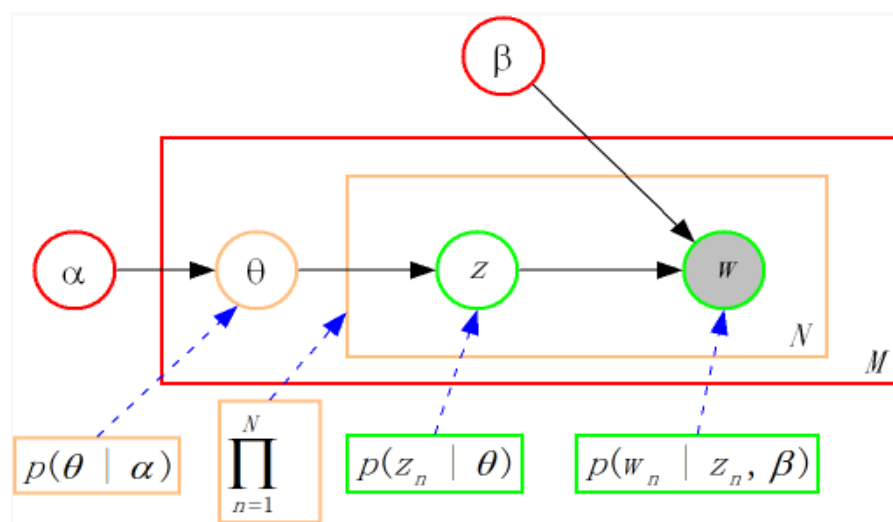


- 主题模型是通过左边矩阵进行训练，学习出右边两个矩阵

在 LDA 模型中，一篇文档生成的方式如下：

- 从狄利克雷分布 α 中取样生成文档 i 的主题分布 θ_i
- 从主题的多项式分布 θ_i 中取样生成文档 i 第 j 个主题 $z_{i,j}$
- 从狄利克雷分布 β 中取样生成主题 $z_{i,j}$ 对应的词语分布 $\phi_{z_{i,j}}$
- 从词语的多项式分布 $\phi_{z_{i,j}}$ 中采样最终生成词语 $w_{i,j}$

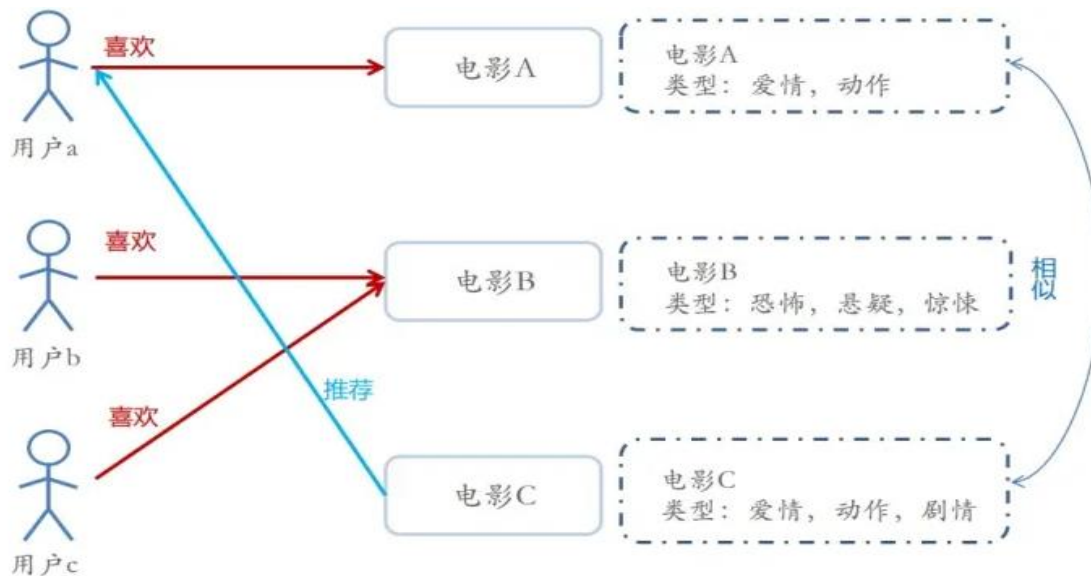
$$p(\theta, \mathbf{z}, \mathbf{w} | \alpha, \beta) = p(\theta | \alpha) \prod_{n=1}^N p(z_n | \theta) p(w_n | z_n, \beta)$$



LDA应用

- **theta文件** (doc—>topic) , **phi文件** (topic—>word)
- 相似文档的计算
- 文本打标签
 - 读取theta文件找出每篇文章概率最大的主题，获取其编号
 - 读取phi文件里每个主题下word概率，读入内存。
 - 找出该文章**最大主题下的最大概率的n个词**作为文章标签。
- 文本推荐：
 - 方法一：基于Topic用户建模的方法
 - 方法二：LDA与LR(逻辑斯蒂回归)进行新闻推荐

协同过滤推荐



	4	?	?	5	1	?	?
	5	5	4				
				2	4	5	
		3					3

基于用户的协同过滤推荐

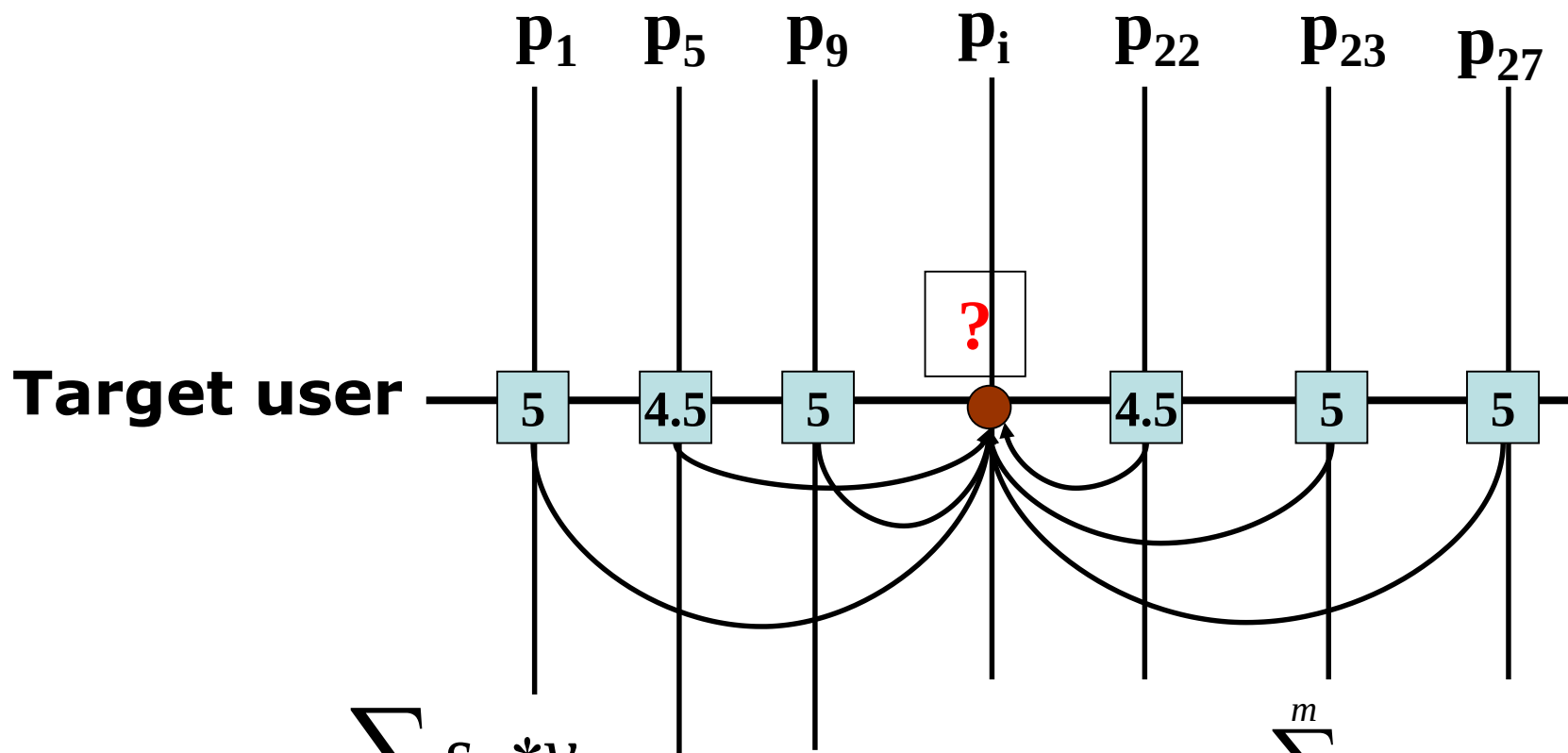
u_5	4	$v_5 = 4$
u_i	?	$v_i = 3.2$
u_8	3	$v_8 = 3.5$
u_9	5	$v_9 = 3$

Users' similarities: $u_{i5} = 0.5$, $u_{i8} = 0.5$, $u_{i9} = 0.8$

$$v_{ij}^* = v_i + K \sum_{v_{kj} \neq ?} u_{ik} (v_{kj} - v_k) \quad u_{ik} = \frac{\sum_j (v_{ij} - v_i)(v_{kj} - v_k)}{\sqrt{\sum_j (v_{ij} - v_i)^2 \sum_j (v_{kj} - v_k)^2}}$$

$$\begin{aligned} v_{ij}^* &= 3.2 + 1/(0.5+0.5+0.8) * [0.5 (4 - 4) + 0.5 (3 - 3.5) + 0.8 (5 - 3)] \\ &= 3.2 + 1/1.8 * [0 - 0.25 + 1.6] = 3.2 + 0.75 = 3.95 \end{aligned}$$

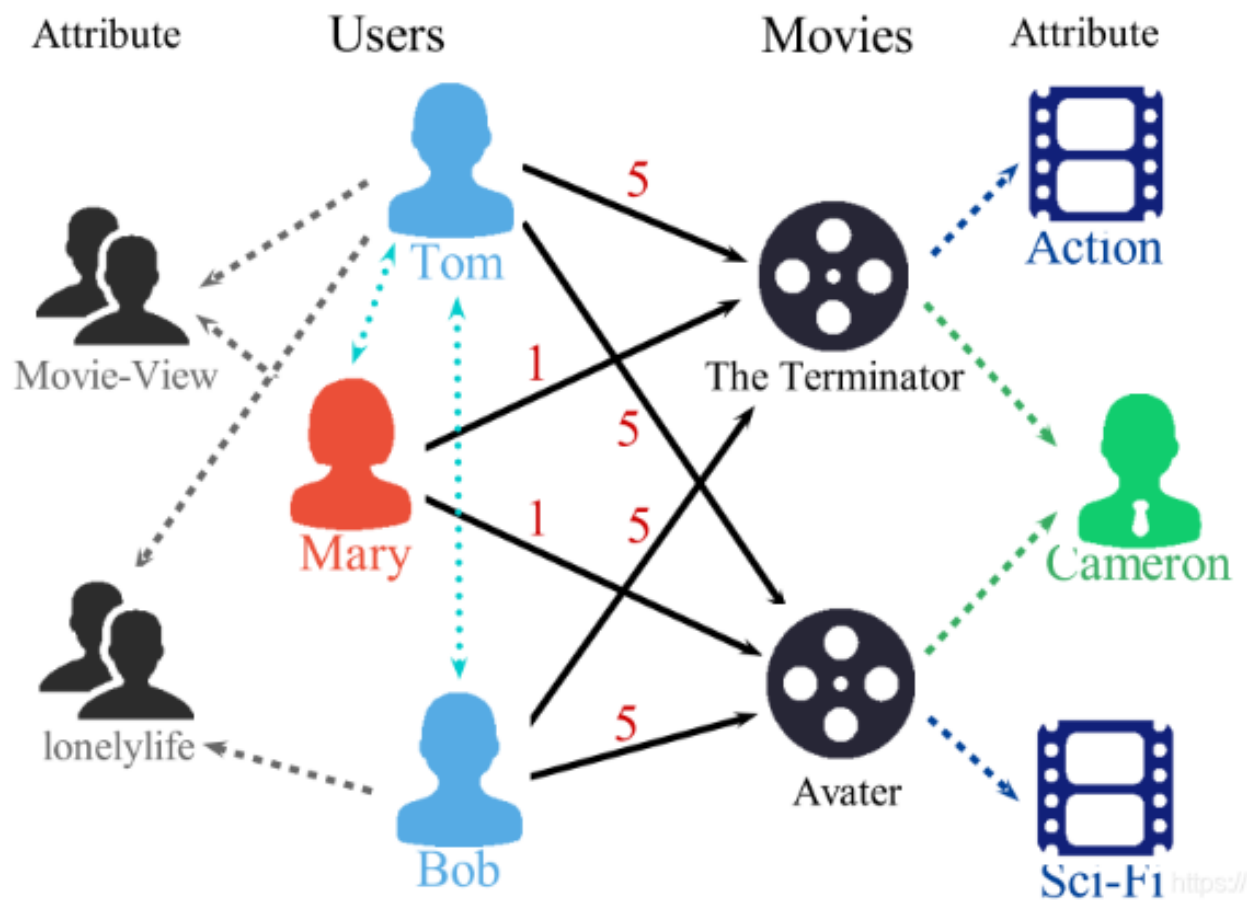
基于项目的协同过滤推荐



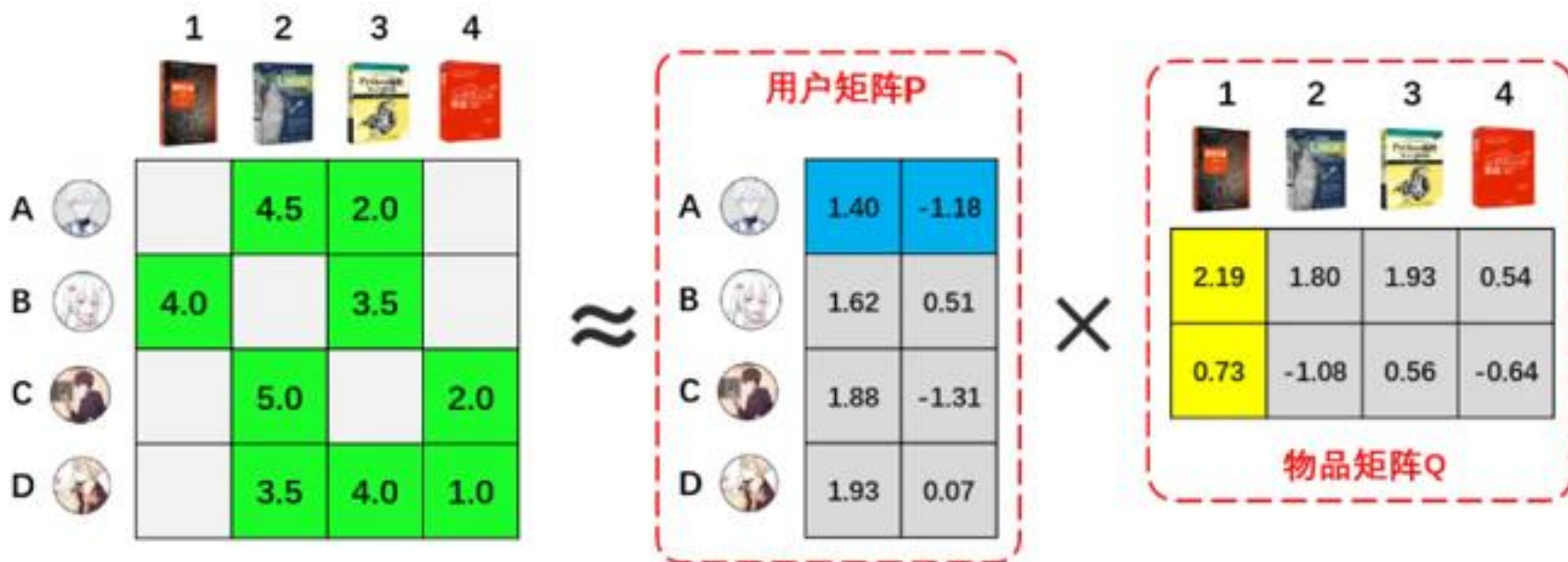
$$v_{ui}^* = \frac{\sum_j s_{ij} * v_{uj}}{\sum_j s_{ij}}$$

$$\cos(p_i, p_j) = \frac{\sum_{k=1}^m v_{ki} v_{kj}}{\sqrt{\sum_{k=1}^m v_{ki}^2 \sum_{k=1}^m v_{kj}^2}}$$

基于图神经网络的推荐



基于模型的推荐（矩阵分解MF）



假如可得到: $R_{U \times I} = P_{U \times K} Q_{K \times I}$

进行预测: $\hat{r}_{ui} = p_u^T q_i$

$$SSE = \sum_{u,i} e_{ui}^2 = \sum_{u,i} \left(r_{ui} - \sum_{k=1}^K p_{uk} q_{ki} \right)^2$$

关键：训练P、Q

$$e_{ui} = r_{ui} - \hat{r}_{ui}$$

$$e_{ui} = r_{ui} - \hat{r}_{ui} = r_{ui} - \mathbf{p}_u^T \mathbf{q}_i = r_{ui} - \sum_{k=1}^K p_{uk} q_{ki}$$

$$SSE = \sum_{u,i} e_{ui}^2 = \sum_{u,i} \left(r_{ui} - \sum_{k=1}^K p_{uk} q_{ki} \right)^2$$

$$\frac{\partial}{\partial p_{uk}} SSE = \frac{\partial}{\partial p_{uk}} (e_{ui}^2) = 2e_{ui} \frac{\partial}{\partial p_{uk}} e_{ui} = -2e_{ui} q_{ki}$$

$$\mathbf{p}_{uk} \dot{=} \mathbf{p}_{uk} - \eta(-e_{ui} \mathbf{q}_{ki}) \dot{=} \mathbf{p}_{uk} + \eta e_{ui} \mathbf{q}_{ki}$$

$$\mathbf{q}_{ki} \dot{=} \mathbf{q}_{ki} - \eta(-e_{ui} \mathbf{p}_{uk}) \dot{=} \mathbf{q}_{ki} + \eta e_{ui} \mathbf{p}_{uk}$$

基于模型的推荐（因子分解机FM）

Feature vector $\mathbf{x}$																			Target y			
$\mathbf{x}^{(1)}$	1	0	0	...	1	0	0	0	...	0.3	0.3	0.3	0	...	13	0	0	0	0	...	5	$y^{(1)}$
$\mathbf{x}^{(2)}$	1	0	0	...	0	1	0	0	...	0.3	0.3	0.3	0	...	14	1	0	0	0	...	3	$y^{(2)}$
$\mathbf{x}^{(3)}$	1	0	0	...	0	0	1	0	...	0.3	0.3	0.3	0	...	16	0	1	0	0	...	1	$y^{(2)}$
$\mathbf{x}^{(4)}$	0	1	0	...	0	0	1	0	...	0	0	0.5	0.5	...	5	0	0	0	0	...	4	$y^{(3)}$
$\mathbf{x}^{(5)}$	0	1	0	...	0	0	0	1	...	0	0	0.5	0.5	...	8	0	0	1	0	...	5	$y^{(4)}$
$\mathbf{x}^{(6)}$	0	0	1	...	1	0	0	0	...	0.5	0	0.5	0	...	9	0	0	0	0	...	1	$y^{(5)}$
$\mathbf{x}^{(7)}$	0	0	1	...	0	0	1	0	...	0.5	0	0.5	0	...	12	1	0	0	0	...	5	$y^{(6)}$
	A	B	C	...	TI	NH	SW	ST	...	TI	NH	SW	ST	...	Time	TI	NH	SW	ST	...		
	User				Movie					Other Movies rated						Last Movie rated						

$$\hat{y}(\mathbf{x}) = w_0 + w_1x_1 + w_2x_2 + \cdots + w_nx_n$$

$$= w_0 + \sum_{i=1}^n w_i x_i,$$

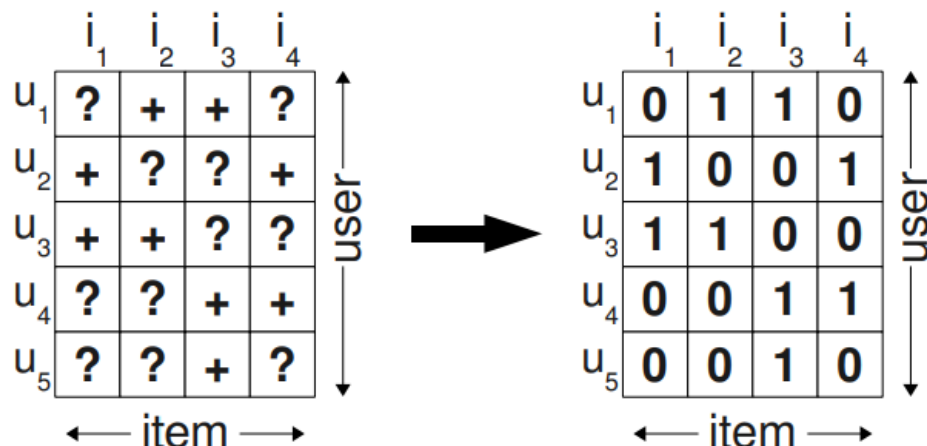
$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle := \sum_{f=1}^k v_{i,f} \cdot v_{j,f}$$

$$y(\mathbf{x}) = \underbrace{w_0 + \sum_{i=1}^n w_i x_i}_{\text{Machine}} + \underbrace{\sum_{i=1}^n \sum_{j=i+1}^n w_{ij} x_i x_j}_{\text{Factorization}}$$

Interaction between variables 

$$\hat{y}(\mathbf{x}) := w_0 + \sum_{i=1}^n w_i x_i + \sum_{i=1}^n \sum_{j=i+1}^n \langle \mathbf{v}_i, \mathbf{v}_j \rangle x_i x_j$$

基于模型的推荐（贝叶斯排序BPR）

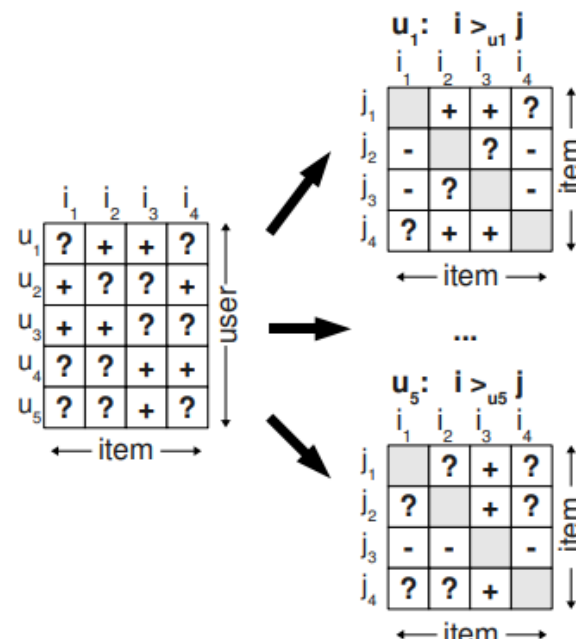


$$p(\Theta | i >_u j) \propto p(i >_u j | \Theta) p(\Theta)$$

$$\prod_{u \in U} p(i >_u j | \Theta) = \prod_{(u, i, j) \in U \times I \times I} p(i >_u j | \Theta)^{\delta((u, i, j) \in D_S)} \cdot (1 - p(i >_u j | \Theta))^{\delta((u, j, i) \in D_S)}$$

$$\delta(b) := \begin{cases} 1 & \text{if } b \text{ is true,} \\ 0 & \text{else} \end{cases}$$

$$p(i >_u j | \Theta) := \sigma(\hat{x}_{uij}(\Theta)) \quad \sigma(x) := \frac{1}{1 + e^{-x}}$$



$$\begin{aligned} \text{BPR-OPT} &:= \ln p(\Theta | i >_u j) \\ &= \ln p(i >_u j | \Theta) p(\Theta) \\ &= \ln \prod_{(u, i, j) \in D_S} \sigma(\hat{x}_{uij}) p(\Theta) \\ &= \sum_{(u, i, j) \in D_S} \ln \sigma(\hat{x}_{uij}) + \ln p(\Theta) \\ &= \sum_{(u, i, j) \in D_S} \ln \sigma(\hat{x}_{uij}) - \lambda_{\Theta} \|\Theta\|^2 \end{aligned}$$



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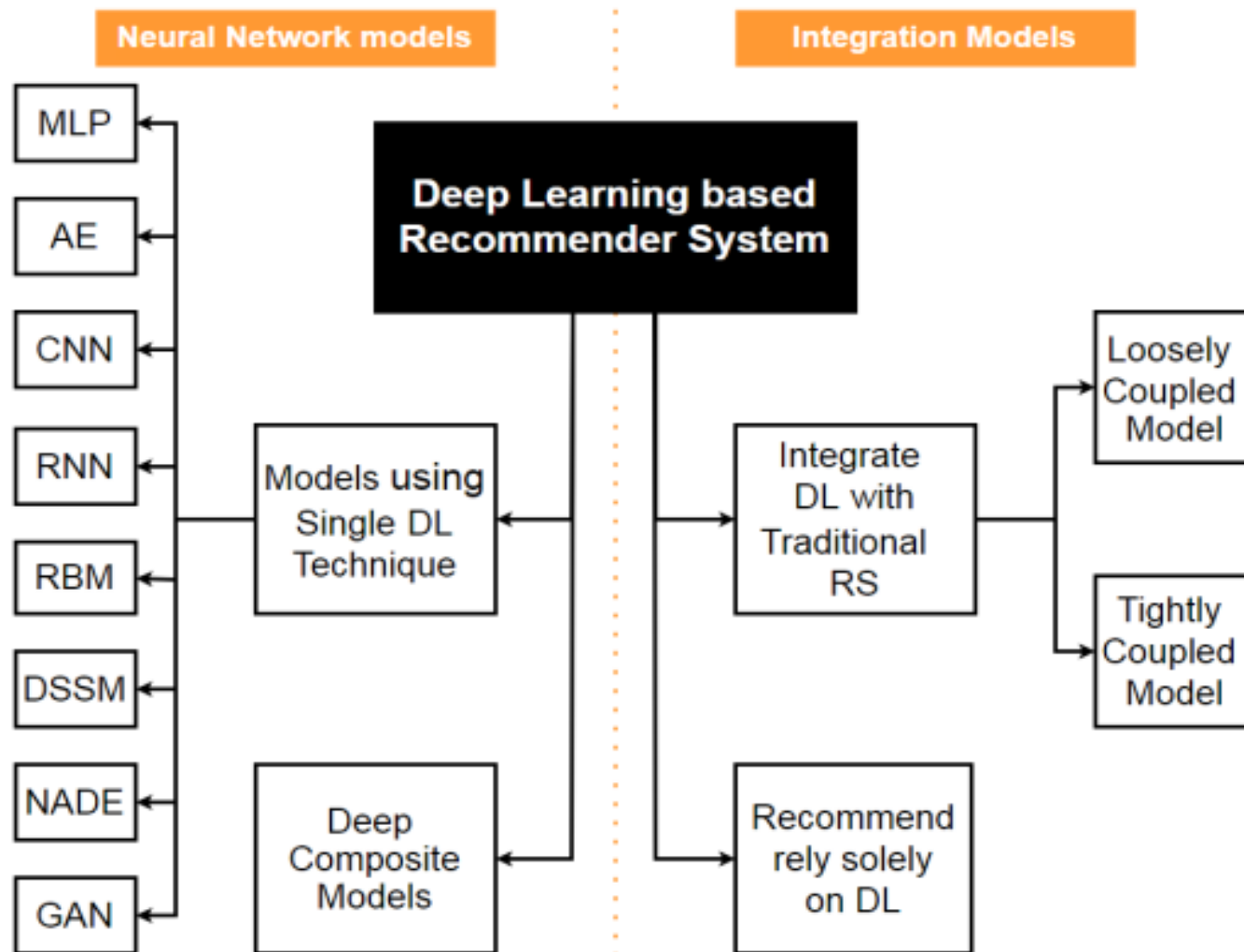


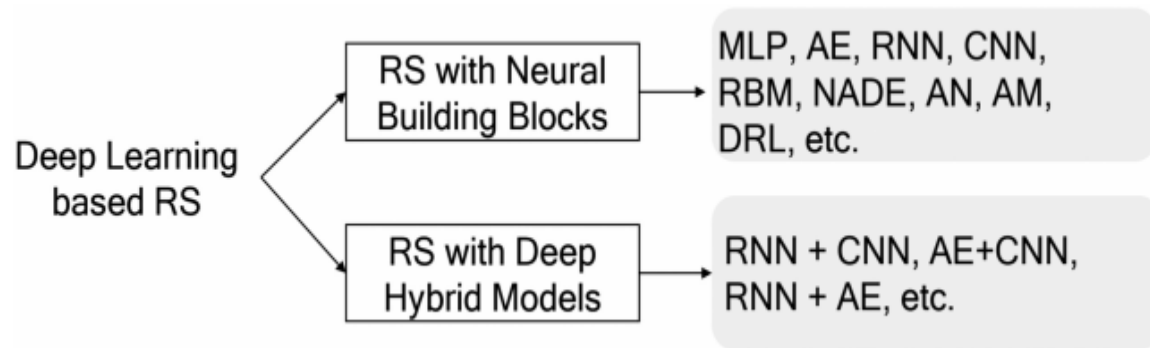
3. 深度学习推荐

推荐系统引入深度学习

- (1) 深度学习特征表示、复杂非线性映射机制；
- (2) 集成各种数据源（文本，图片，视频，音频等）通过提取更高质量的特征表示来丰富推荐算法用户偏好建模；
- (3) 深度学习有可能对用户偏好进行直接建模。不再依赖于协同过滤，新用户的冷启动问题得到了解决；
- (4) 为推荐系统中的关键问题提供了丰富的解决方案，传统的机器学习限制矩阵分解MF、LDA中，深度学习DAE、RNN、LSTM等广泛应用。

Deep Learning based Recommender System





Categories	Publications
MLP	[2, 13, 20, 27, 38, 47, 53, 54, 66, 92, 95, 158, 167, 186], [12, 39, 93, 112, 135, 155, 183, 184]
Autoencoder	[34, 88, 89, 114, 116, 126, 137, 138, 141, 160, 178, 188, 208], [4, 10, 32, 94, 151, 152, 159, 171, 172, 189, 197, 209, 210]
CNNs	[25, 49, 50, 75, 76, 99, 105, 128, 131, 154, 166, 173, 203, 207], [6, 44, 51, 83, 110, 127, 144, 149, 170, 191, 192]
RNNs	[5, 28, 35, 56, 57, 73, 78, 90, 118, 133, 140, 143, 175–177], [24, 29, 33, 55, 68, 91, 108, 113, 134, 142, 150, 174, 180]
RBM	[42, 71, 72, 101, 124, 168, 181]
NADE	[36, 204, 205]
Neural Attention	[14, 44, 70, 90, 100, 103, 128, 146, 170, 190, 195, 206], [62, 147, 194]
Adversary Network	[9, 52, 163, 165]
DRL	[16, 21, 107, 169, 199–201]
Hybrid Models	[17, 38, 41, 82, 84, 87, 119, 136, 161, 193, 194]

Data Sources/Tasks	Notes
Sequential Information	w/t User ID
	Session based w/o User ID
	Check-In, POI
Text	Hash Tags
	News
	Review texts
	Quotes
Images	Visual features
Audio	Music
Video	Videos
Networks	Citation Network
	Social Network
	Cross Domain
Others	Cold-start
	Multitask
	Explainability

Deep Matrix Factorization(DMF) (IJCAI2017)

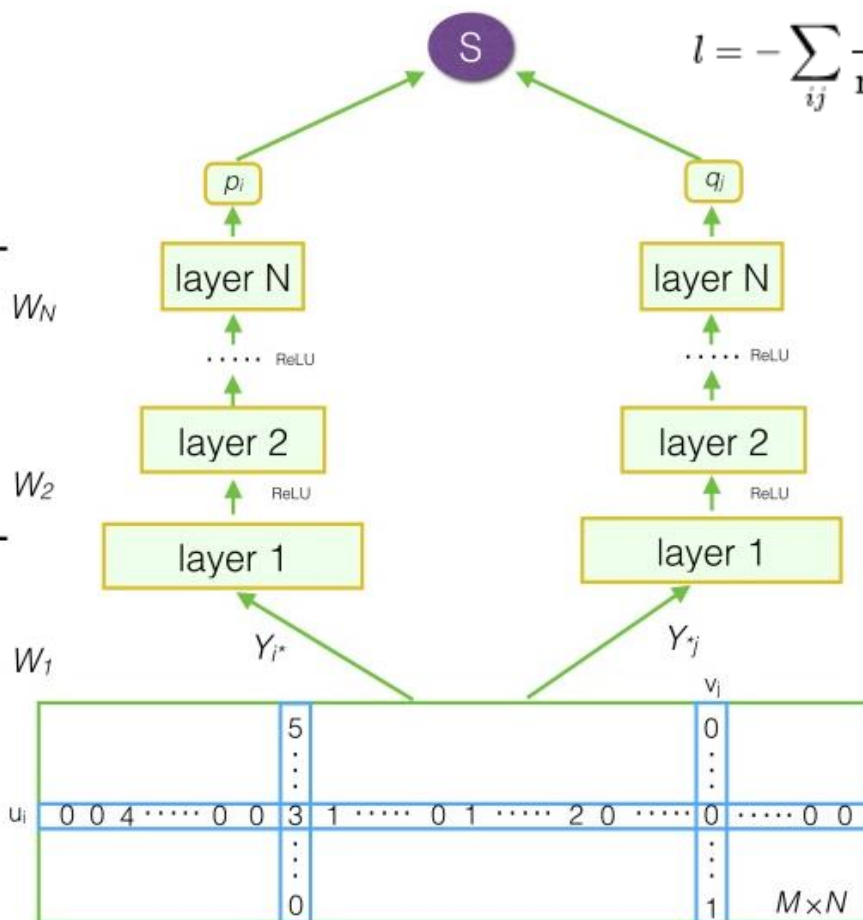
$$\hat{Y}_{ij} = F^{DMF}(u_i, v_j | \Theta) = \text{cosine}(p_i, q_j) = \frac{p_i^T q_j}{\|p_i\| \|q_j\|}$$

Relevance measured
by cosine similarity

$$l = - \sum_{ij} \frac{Y_{ij}}{\max(R)} \log \hat{Y}_{ij} + (1 - \frac{Y_{ij}}{\max(R)}) \log(1 - \hat{Y}_{ij})$$

Latent representation

Multi-layer non-
linear projection

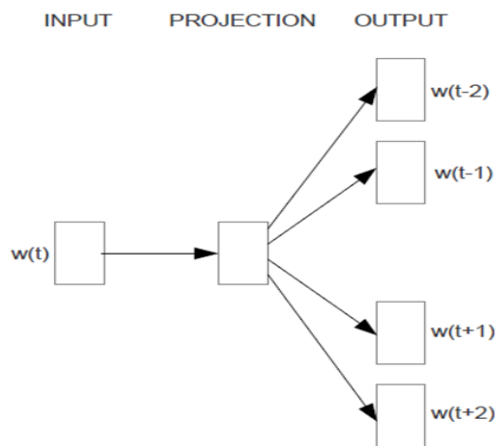
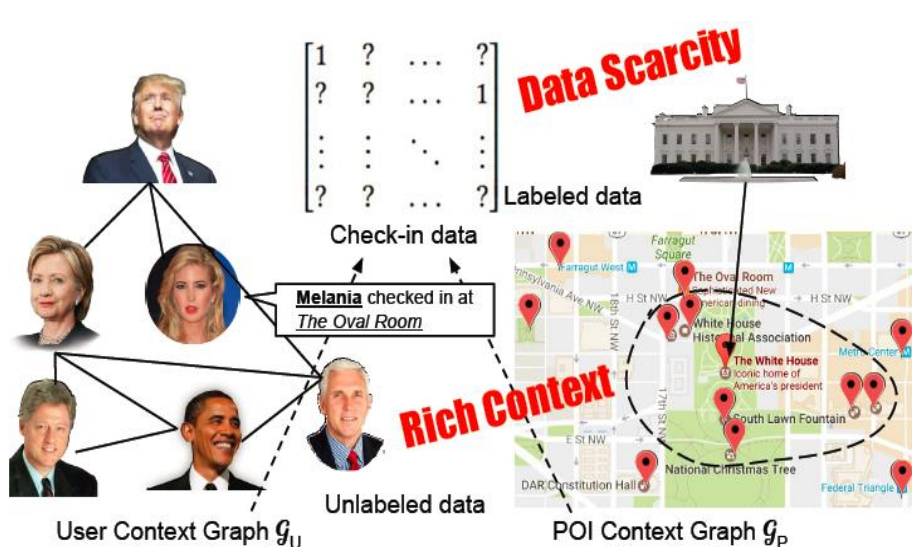


$$p_i = f_{\theta_N^U}(\dots f_{\theta_3^U}(W_{U2} f_{\theta_2^U}(Y_{i*} W_{U1})) \dots)$$

$$q_j = f_{\theta_N^I}(\dots f_{\theta_3^I}(W_{V2} f_{\theta_2^I}(Y_{*j}^T W_{V1})) \dots)$$

$$\begin{aligned} l_1 &= W_1 x \\ l_i &= f(W_{i-1} l_{i-1} + b_i), i = 2, \dots, N-1 \\ h &= f(W_N l_{N-1} + b_N) \end{aligned}$$

Preference and Context Embedding(PACE) (KDD2017)

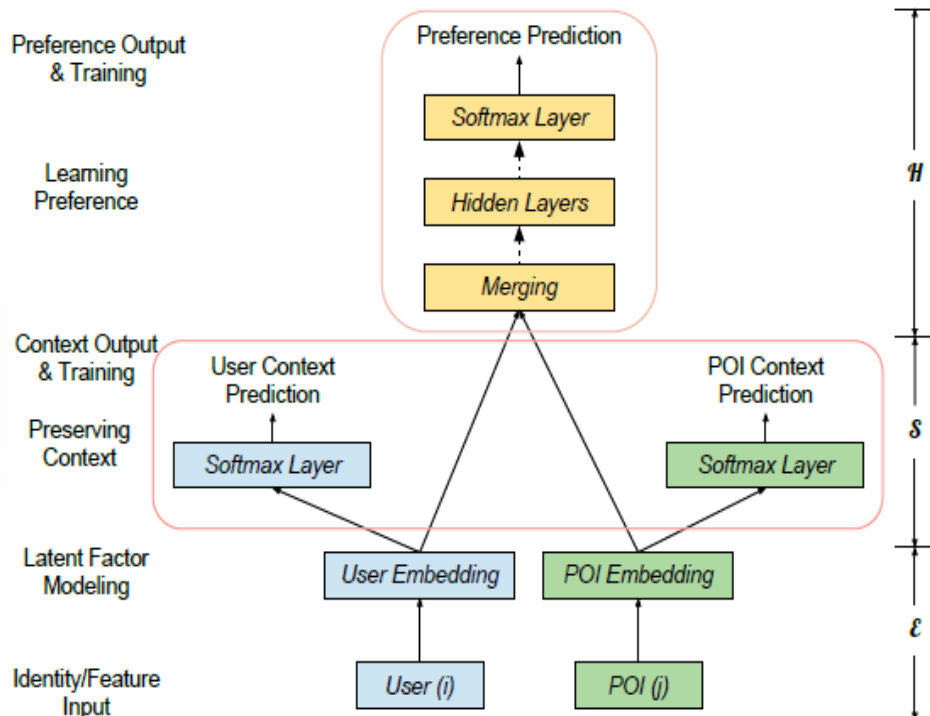


Skip-gram

一个用户用
他周围的朋友来表示

多信息集成

多目标优化



$$\begin{aligned}\mathcal{J}_P &= \log p(\mathcal{L} | \Theta_e, \Theta_h) \\ &= - \sum_{(u_i, p_j) \in \mathcal{L}^+} \log \hat{y}_{ij} - \sum_{(u_i, p_j) \in \mathcal{L}^-} \log (1 - \hat{y}_{ij}) \\ &= - \sum_{(u_i, p_j) \in \mathcal{L}} y_{ij} \log \hat{y}_{ij} + (1 - y_{ij}) \log (1 - \hat{y}_{ij})\end{aligned}$$

$$\mathcal{J} = \mathcal{J}_P + \lambda_1 \mathcal{J}_{C_u} + \lambda_2 \mathcal{J}_{C_p}$$

Deep Matrix Factorization(DMF) (IJCAI2017)

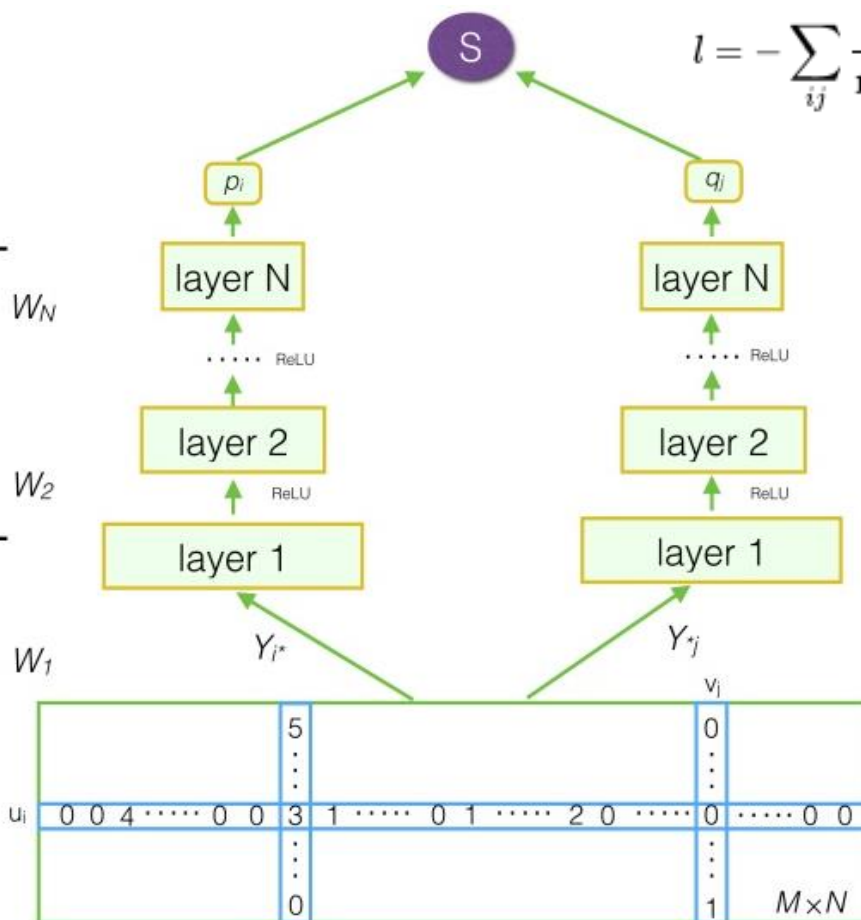
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$$l = - \sum_{ij} \frac{Y_{ij}}{\max(R)} \log \hat{Y}_{ij} + (1 - \frac{Y_{ij}}{\max(R)}) \log(1 - \hat{Y}_{ij})$$

Latent representation

Multi-layer non-
linear projection



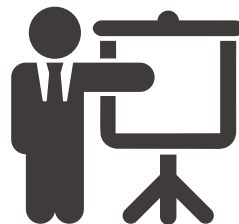
$$p_i = f_{\theta_N^U}(\dots f_{\theta_3^U}(W_{U2} f_{\theta_2^U}(Y_{i*} W_{U1})) \dots)$$

$$q_j = f_{\theta_N^I}(\dots f_{\theta_3^I}(W_{V2} f_{\theta_2^I}(Y_{*j}^T W_{V1})) \dots)$$

$$\begin{aligned} l_1 &= W_1 x \\ l_i &= f(W_{i-1} l_{i-1} + b_i), i = 2, \dots, N-1 \\ h &= f(W_N l_{N-1} + b_N) \end{aligned}$$



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4. 高级推荐系统

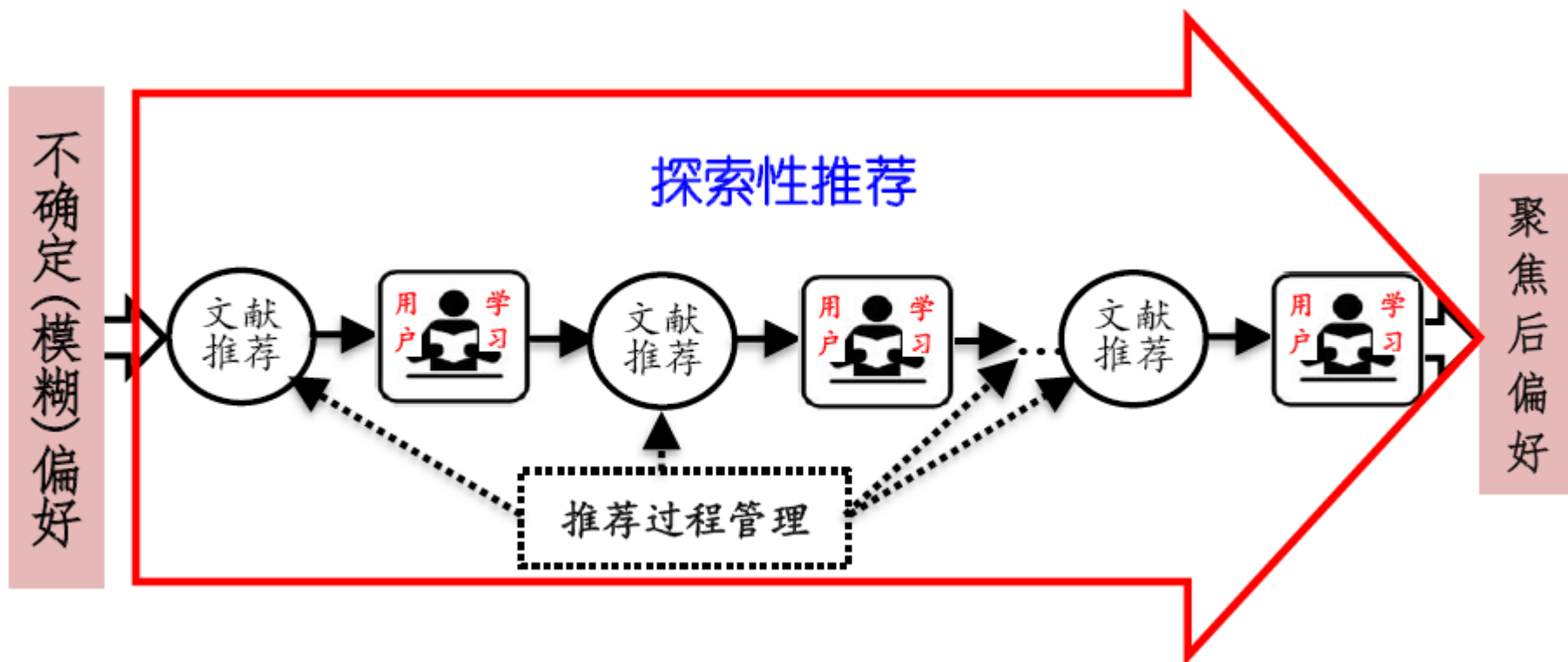
why

- 用户偏好清楚
 - 能表达
 - 不能表达 (太复杂、不断变化)
- 用户偏好不清晰
- 有责任帮助用户挖掘需求

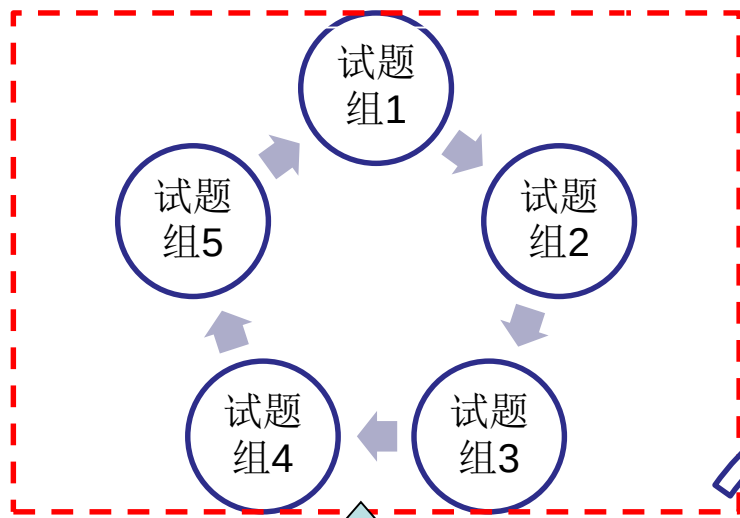
探索性推荐

■ 推荐依据

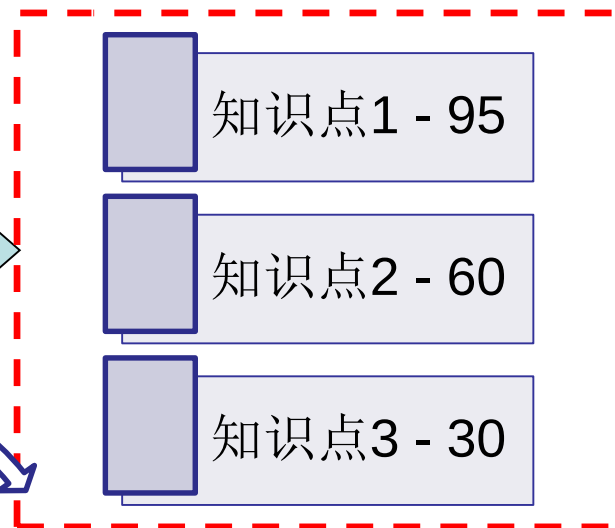
- 基于偏好？探索性质？



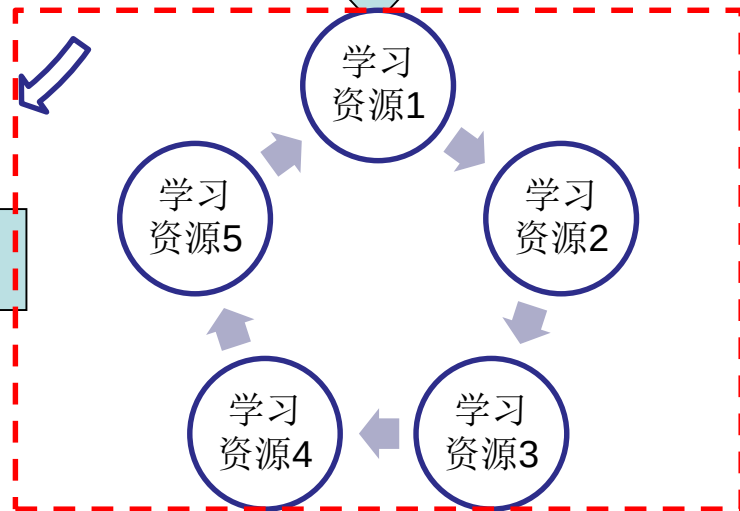
自适应测试



精准化评估



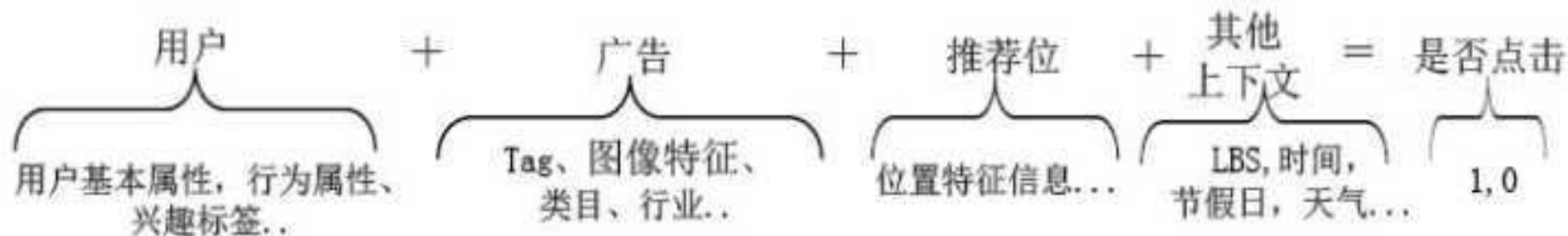
针对性学习



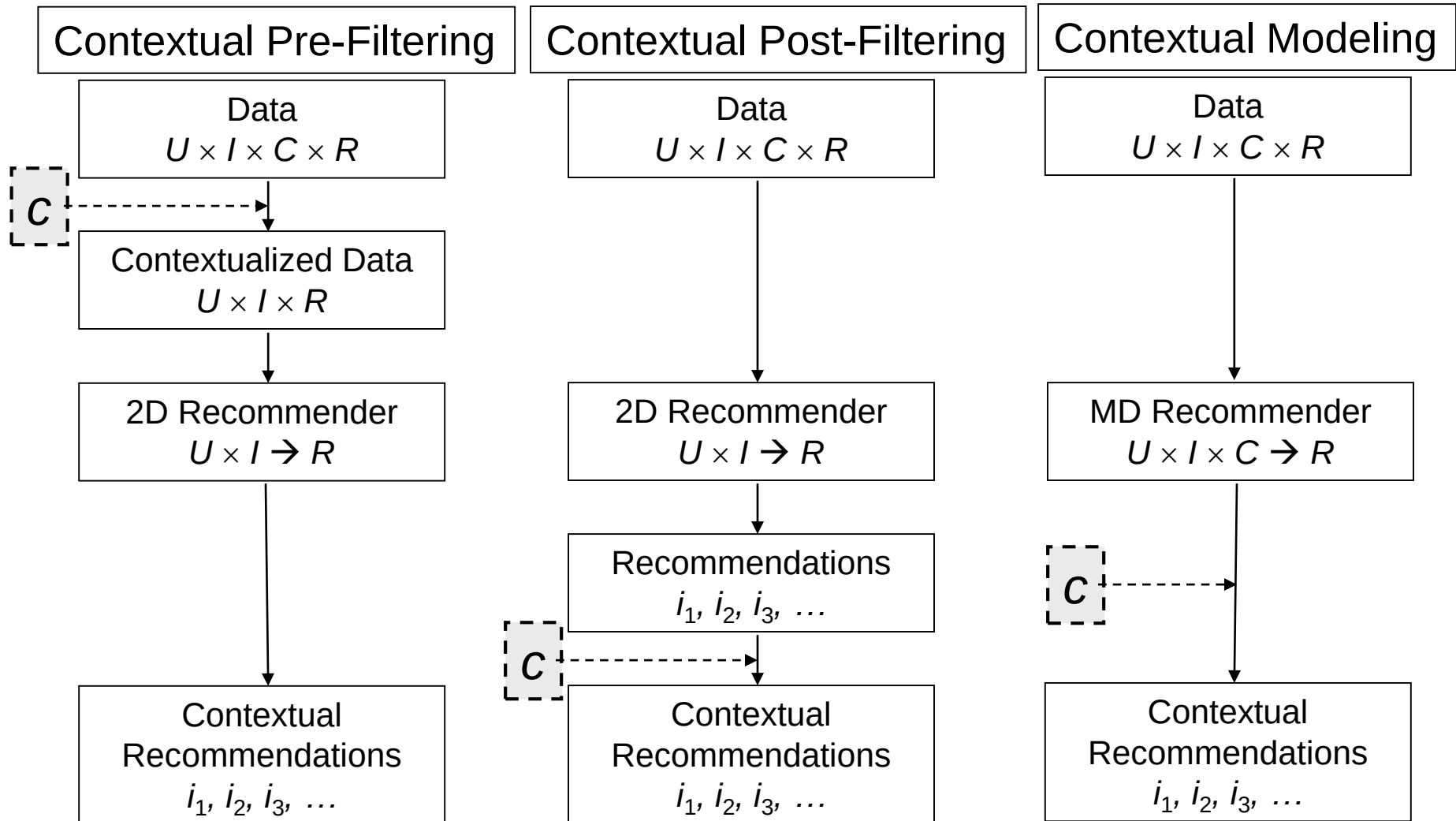
个性化推荐

Context Dependent Recommendations

- Recommend a vacation
 - Winter vs. summer
- Recommend a purchase (e-retailer)
 - Gift vs. for yourself
- Recommend a movie
 - To a student who wants to see it on Saturday night with his girlfriend in a movie theater.

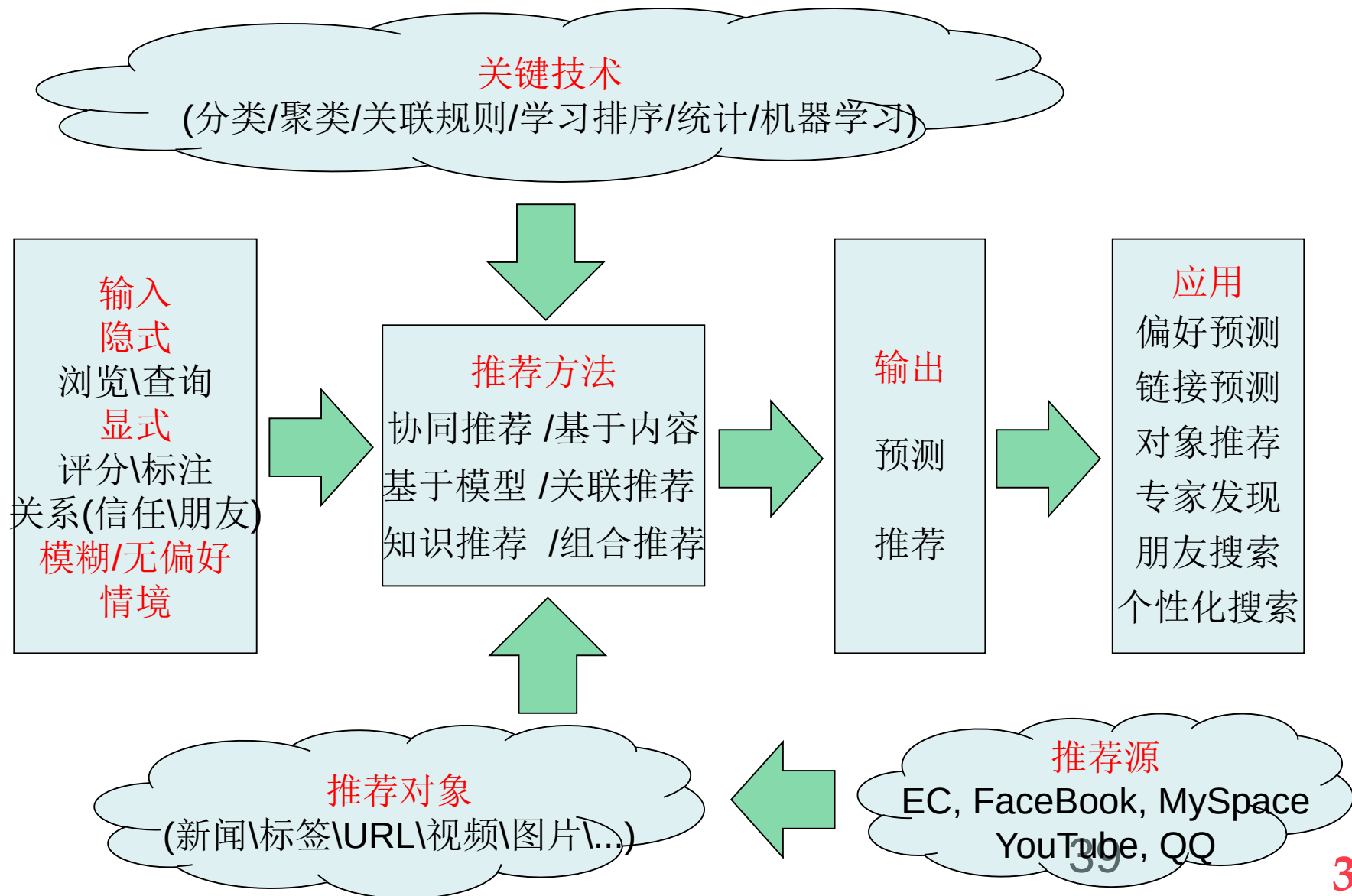


Incorporating Context in RecSys



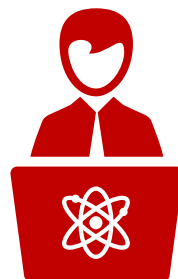
[Adomavicius and Tuzhilin 2008]

Conclusions





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谢谢大家！

